# 25PY101: Engineering Physics

Dr. Sreekar Guddeti

December 4, 2025

## References

Module 2 pre-T1 slot2

---

# 1 Topics

| Topics | Avadhanulu | Neamen |
|---|---|---|
| $E_F$ in ext. s.c. | | |
| | 37.15 Ext s.c. | 4.2 Dopant atoms and energy levels |
| | 37.16 n-type | 4.3.1 Equilibrium electron and hole concentration |
| | 37.17 p-type | |
| Band diagrams of n-type and p-type | | |
| | 37.16 Band diagrams @ $T = 0\,\mathrm{K}$ and $T \neq 0\,\mathrm{K}$ | 4.4.2 Complete ionization and freeze out |
| | 37.23 $E_F$ in ext. s.c. | 4.6 Position of Fermi level |
| pn junction diode – Forward and reverse bias | | |
| | 38.2 pn junction diode | 7.1 Basic structure of pn junction |
| | 38.3 pn junction diode under forward bias | 7.2 Zero applied bias |
| | 38.4 pn junction diode under reverse bias | 7.3 Reverse applied bias |
| | 38.6 $I - V$ characteristic | 8.1.1 Qualitative description of charge in pn junction |
| | | 8.1.5 Ideal pn junction current |
| Solar cell | | |
| | 38.12 Solar cell | 14.2.1 pn junction solar cell |
| | | 14.2.2 Conversion efficiency and solar concentration |

# 2 Test your understanding

## $E_F$ in extrinsic semiconductor

1. What is a donor impurity? What is an acceptor impurity?
2. Differentiate between n-type and p-type semiconductor by sketching their band diagrams.
3. p-type and n-type have more holes and electrons respectively. Explain why they are electrically neutral.
4. What is meant by complete ionization? What is meant by freeze-out?
5. What is the effect of temperature on the working of n-type and p-type semiconductors? Draw energy band diagrams to show the variation in the position of Fermi level with rise in temperature.

6. Draw enery band diagrams for n-type semiconductor at 0 K and at room temperature.
7. Distinguish between extrinsic and intrinsic semiconductors.
8. Sketch a graph of $n_0$ versus temperature for an n-type semiconductor.
9. Sketch a graph of $E_F$ versus temperature for p-type semiconductor.

### pn junction diode

1. What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined?
2. Explain the formation of **potential barrier** across the pn junction region.
3. Explain the formation of **depletion region** in the pn junction region.
4. Draw band diagrams for symmetrically doped pn junction at
   - zero bias
   - forward bias
   - reverse bias
5. What is reverse saturation current in a diode?
6. Why is reverse saturation current independent of reverse bias?
7. What is reverse breakdown?

### Solar cell

1. Explain the working of solar cell.
2. Draw and explain the $I - V$ characteristics of a solar cell.

## 3 Technical terms

### $E_F$ in extrinsic semiconductor

- acceptor atoms
- donor atoms
- extrinsic semiconductor
- complete ionization
- freeze-out

### pn junction diode

- built in potential barrier
- depletion region
- space charge region
- space charge width
- metallurgical junction

## Solar cell

- conversion efficiency
- fill factor
- open-circuit voltage
- short-circuit current